Exponential randomized-time lower bound for randomized Kt complexity and its gap version

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Editorial note. Faithful editable transcription of complete author Section5 Theorems5.3/5.8 and proofs, including exact prior reconstruction tool, author Corollary5.5/Lemma5.6 support and full page21 gap/i.o. continuation. Exact and gap variants count once. Original author mathematical expressions retained; no new proof or correction. Original CC BY4.0 license and complete bibliography retained.

Allowed standard prerequisites (editorial).

- Success-probability robustness of randomized Kt. Exact prior Lu-Oliveira Lemma5.2 retained.
- Trevisan/Raz-Reingold-Vadhan reconstructive PRG construction. Author exact Lemma5.4 citing prior [32, Theorem4.7] and original Proof Sketch retained; this is a standard construction premise, not a replaced main proof.

3. Preliminaries: Notation. $[n]$ denotes $\{1, \dots, n\}$. For a function $p : \mathbb{N} \to \mathbb{N}$ and $i \in \mathbb{N}$, the function $p^{(i)} : \mathbb{N} \to \mathbb{N}$ is recursively defined as follows: $p^{(0)}(n) := n$ and $p^{(i+1)}(n) := p^{(i)}(p(n))$ for every $n \in \mathbb{N}$.

Kolmogorov Complexity. For a string $x \in \{0,1\}^*$ and an index $i \in \mathbb{N}$, let x_i denote the i-th bit of x if $i \leq |x|$ and $\perp$ if $i > |x|$. We fix an efficient universal Turing machine U. Time-bounded Kolmogorov complexity is formally defined as follows.

Definition 3.1 (Time-bounded Kolmogorov complexity). For strings $x, y \in \{0,1\}^*$, a time bound $t \in \mathbb{N} \cup \{\infty\}$, and an oracle A, the A-oracle t-time-bounded Kolmogorov complexity of x given y is defined as

$$K^{t,A}(x \mid y) := \min\{|d| : U^{A,y,d} \text{ outputs } x_i \text{ on input } i \text{ in time } t \text{ for every } i \in [1, \dots, |x|+1]\}.$$

Here, $U^{A,y,d}$ indicates that the universal Turing machine is given oracle access to A and each bit of y and $d \in \{0,1\}^*$. We omit the superscript A if $A = \varnothing$, the superscript t if $t = \infty$, and "$\mid y$" if y is the empty string. A simple counting argument implies the following basic fact of Kolmogorov-randomness.

Fact 3.2. For any integer $s \geq 1$ and any string $y \in \{0,1\}^*$, the number of strings $x \in \{0,1\}^*$ such that $K(x \mid y) < s$ is less than 2^s.

Proof. The number of programs of length less than s is at most $\sum_{i=0}^{s-1} 2^i < 2^s$. □

5. A Lower Bound for Randomized Kt Complexity. Here, we implement Kolmogorov's insightful approach to the P versus NP problem for randomized Kt complexity. We recall the formal definition of rKt, which was introduced by Oliveira [62].

Definition 5.1 ([62]). For every $x \in \{0,1\}^*$ and every $\lambda \in [0,1]$, the randomized time-bounded Kolmogorov complexity $rKt_\lambda(x)$ of x is defined as

$$rKt_\lambda(x \mid y) := \min\left\{ |d| + \lceil \log t \rceil : \Pr_{r \sim \{0,1\}^t}[U(d,y,r) \text{ outputs } x \text{ in time } t] \geq \lambda \right\}.$$

We omit the subscript "λ" if $\lambda = 2/3$ and "$\mid y$" if y is the empty string. The language MrKtP is defined as

$$\mathsf{MrKtP} := \{(x, 1^s) : rKt(x) \leq s\}.$$

The definition of rKt_λ is robust with respect to a choice of λ.

Lemma 5.2 (Lu and Oliveira [59]). For all $x \in \{0,1\}^*$ and $\lambda \in (0,1]$,

$$rKt(x) \leq rKt_\lambda(x) + O(\log(1/\lambda)).$$

In what follows, let $\epsilon > 0$ be a sufficiently small positive constant. The O notation below does not depend on ϵ. We say that a pseudo-deterministic algorithm M computes y on input x [27] if

$$\Pr_M[M(x) = y] \geq 2/3,$$

where the probability is taken over an internal coin flip sequence of the randomized algorithm M. We use the following pseudorandom generator construction.

Lemma 5.4 (see the proof of [32, Theorem 4.7]). For all sufficiently large $n, m \in \mathbb{N}$ such that $m \leq 2n$, there exists a triple $(G, A, R^{(-)})$ such that

$$\begin{aligned} G &: \{0,1\}^n \times \{0,1\}^d \to \{0,1\}^m, \\ A &: \{0,1\}^n \times \{0,1\}^d \to \{0,1\}^m, \\ R^{(-)} &: \{0,1\}^m \times \{0,1\}^d \times \{0,1\}^r \to (\{0,1\}^n)^L, \end{aligned}$$

and for every $x \in \{0,1\}^n$ and any $D : \{0,1\}^m \to \{0,1\}$ such that

$$\left| \Pr_{\substack{z \sim \{0,1\}^d \\ w' \sim \{0,1\}^{O(m)}}} [D(G(x;z);w') = 1] - \Pr_{\substack{w \sim \{0,1\}^m \\ w' \sim \{0,1\}^{O(m)}}} [D(w;w') = 1] \right| \geq \frac{1}{m},$$

it holds that

$$\Pr_{\substack{w \sim \{0,1\}^r \\ z \sim \{0,1\}^d}} [x \in R^D(A(x,z),z,w)] \geq \frac{1}{2m^2}.$$

Here, we have $d = O(\log^3 n)$, $r = O(m)$, and $L = \mathrm{poly}(m)$. Moreover, G and A can be computed in time $\mathrm{poly}(n)$ and R^D can be computed in time $\mathrm{poly}(n)$ with oracle access to D.

Proof Sketch. The construction is given by the improvement of Trevisan's extractor [70] given by Raz et al. [64]. Specifically, we encode x using a list-decodable error-correcting code and regard the encoding of x as the truth table of a hard function f. Then, the pseudorandom generator construction G is defined as the Nisan–Wigderson generator [61] instantiated with the function f and the weak design of [64]. ([32, Theorem 4.7] is stated as a black-box hitting set generator construction; however, it is easy to observe that the construction actually provides a pseudorandom generator construction, which is a stronger object.) □

Theorem 5.3. There exists a constant $\epsilon > 0$ such that $\mathsf{MrKtP} \notin \mathsf{BPTIME}(2^{\epsilon n})$.

Theorem 5.8 (Theorem 1.4, restated). $\mathsf{GapMrKtP} \notin \mathsf{i.o.BPTIME}(2^{\epsilon n})$ for some constant $\epsilon > 0$.

Corollary 5.5. Under the same hypothesis of Lemma 5.4, it holds that

$$rKt^D(x) \leq m + O(\log^3 n).$$

Proof. By an averaging argument, there exists a string $z \in \{0,1\}^d$ such that

$$\Pr_w [x \in R^D(A(x,z),z,w)] \geq \frac{1}{2m^2}.$$

Let $\alpha := A(x,z) \in \{0,1\}^m$. The string x can be described by the following pseudo-deterministic algorithm M: M takes the advice string $\alpha \in \{0,1\}^m$ and $z \in \{0,1\}^d$ as input, picks $w \sim \{0,1\}^r$ randomly, and outputs a random element of $R^D(\alpha,z,w)$. Since the list size is at most $L = \mathrm{poly}(m)$, the probability that the output of M is equal to x is at least $\lambda := \frac{1}{2m^2} \cdot \frac{1}{L}$. Therefore, we obtain $rKt_\lambda(x) \leq m + d + O(\log n)$. The result follows from Lemma 5.2. □

Lemma 5.6. If $\mathsf{MrKtP} \in \mathsf{BPTIME}(2^{\epsilon n})$, then for all sufficiently large $n \in \mathbb{N}$ and any strings $x \in \{0,1\}^n$ and $y \in \{0,1\}^n$, it holds that

$$rKt(x \mid y) + rKt(y) \leq rKt(x,y) + O(\epsilon n).$$

Proof. We may assume without loss of generality that $rKt(y) \gg \log^3 n$. Let m and m' be parameters chosen later. Let $G : \{0,1\}^n \times \{0,1\}^d \to \{0,1\}^m$ and $G' : \{0,1\}^n \times \{0,1\}^{d'} \to \{0,1\}^{m'}$ be the pseudorandom generator construction of Lemma 5.4 with parameters m and m', respectively. Fix $z \in \{0,1\}^d$ and $z' \in \{0,1\}^{d'}$. Since the strings $(G(x;z), G'(y;z'))$ can be described by z, z', and a program that computes (x,y), we obtain

$$\begin{aligned} rKt(G(x;z), G'(y;z')) &\leq rKt(x,y) + |z| + |z'| + O(\log n) \\ &= rKt(x,y) + O(\log^3 n). \end{aligned} \quad (13)$$

Pick $w \sim \{0,1\}^m$ and $z' \sim \{0,1\}^{d'}$ randomly. We claim that $rKt(w, G'(y;z')) \geq |w| + |w'| - O(\log n)$ with high probability. Toward a contradiction, assume that

$$\Pr_{w,z'}[rKt(w, G'(y;z')) < |w| + |w'| - O(\log n)] \geq \frac{1}{2}.$$

By Fact 3.2, we have

$$\Pr_{w,w'}[rKt(w,w') \geq |w| + |w'| - O(\log n)] \geq 1 - o(1).$$

We define an oracle D so that $D(w';w) := 1$ if and only if $rKt(w,w') < |w| + |w'| - O(\log n)$. Then, the two inequalities above show that D distinguishes the output distribution of $G'(y;-)$ from the uniform distribution. By Corollary 5.5, we obtain

$$rKt^D(y) \leq m' + O(\log^3 n).$$

Using the assumption that $\mathsf{MrKtP} \in \mathsf{BPTIME}(2^{\epsilon n})$, the oracle D can be computed by a randomized algorithm running in time $2^{O(\epsilon m')}$. It follows that

$$rKt(y) \leq m' + O(\epsilon m') = (1 + O(\epsilon)) \cdot m'.$$

We choose m' so that this inequality does not hold; that is,

$$m' := (1 - O(\epsilon)) \cdot rKt(y).$$

Then, we obtain a contradiction and conclude that

$$\Pr_{w,z'}[rKt(w, G'(y;z')) \geq m + m' - O(\log n)] \geq \frac{1}{2}. \quad (14)$$

Define D_y to be an oracle such that $D_y(w;z') := 1$ if and only if $rKt(w, G'(y;z')) \leq rKt(x,y) + O(\log^3 n)$. By Equations (14) and (13), D_y distinguishes the output distribution of $G(x;-)$ from the uniform distribution if

$$m + m' - O(\log n) \geq rKt(x,y) + O(\log^3 n).$$

We define $m := rKt(x,y) - m' + O(\log^3 n)$ so that this inequality holds. By Corollary 5.5, we obtain

$$rKt^{D_y}(x) \leq m + O(\log^3 n).$$

Since the oracle D_y can be computed in time $2^{O(\epsilon m)}$ given y as hard-wired input, we conclude that

$$\begin{aligned} rKt(x \mid y) &\leq m + O(\epsilon m). \\ &\leq rKt(x, y) - m' + O(\epsilon n). \\ &\leq rKt(x, y) - (1 - O(\epsilon)) \cdot rKt(y) + O(\epsilon n). \\ &\leq rKt(x, y) - rKt(y) + O(\epsilon n). \end{aligned}$$

□

Proof of Theorem 5.3. The outline of the proof is as follows: First, we find a string $y \in \{0,1\}^n$ such that $rKt(y) \geq n$ by an exhaustive search over all strings $y \in \{0,1\}^n$. Next, we find a string $x \in \{0,1\}^n$ such that $rKt(x, y) \geq 2n - O(\epsilon n)$ by an exhaustive search over all strings $x \in \{0,1\}^n$. The existence of such a string x is guaranteed by Lemma 5.6. Finally, we observe that the pair (x, y) can be pseudo-deterministically computed in time $2^{n+O(\epsilon n)}$, which contradicts the lower bound on $rKt(x, y)$. Details follow. Using the assumption that $\mathsf{MrKtP} \in \mathsf{BPTIME}(2^{\epsilon n})$, let M be the randomized algorithm that computes $rKt(y)$ on input y of length n in time $2^{O(\epsilon n)}$ with probability at least $1 - 2^{-2n}$. Fix $n \in \mathbb{N}$. Let y_n be the lexicographically first string $y \in \{0,1\}^n$ such that $rKt(y) \geq n$. Note that the existence of such a string y_n is guaranteed by Fact 3.2. We claim that there exists a pseudo-deterministic algorithm M_1 that computes y_n on input $n \in \mathbb{N}$ in time $2^{n+O(\epsilon n)}$. For all strings $y \in \{0,1\}^n$ in lexicographical order, the algorithm M_1 tests whether $rKt(y) \geq n$ using M, and outputs the first string y that passes this test. Since the error probability of M is at most 2^{-2n}, by a union bound, the algorithm M_1 computes y_n pseudo-deterministically with probability $1 - 2^{-n}$. The running time of M_1 is at most $2^{n+O(\epsilon n)}$. Next, let x_n be the lexicographically first string $x \in \{0,1\}^n$ such that

$$rKt(x, y_n) \geq 2n - c \cdot \epsilon n, \tag{15}$$

where c is a large constant to be chosen independently of ϵ. We prove that x_n is well defined, i.e., there exists a string $x \in \{0,1\}^n$ such that $rKt(x, y_n) \geq 2n - c \cdot \epsilon n$. By Fact 3.2, there exists a string $x \in \{0,1\}^n$ such that $rKt(x \mid y) \geq n$. By Lemma 5.6,

$$rKt(x, y) + O(\epsilon n) \geq rKt(x \mid y) + rKt(y) \geq 2n.$$

Choosing a large constant c, we obtain $rKt(x, y) \geq 2n - c \cdot \epsilon n$, as desired. We claim that there exists a pseudo-deterministic algorithm M_2 that computes x_n given y_n as input in time $2^{n+O(\epsilon n)}$. For all strings $x \in \{0,1\}^n$ in lexicographical order, the algorithm M_2 tests whether $rKt(x, y_n) \geq 2n - c \cdot \epsilon n$ using M, and outputs the first string x that passes this test. Using a union bound, the error probability of M_2 is bounded by 2^{-n}. The running time of M_2 is at most $2^{n+O(\epsilon n)}$. Finally, by combining the algorithms M_1 and M_2, we obtain an algorithm that computes (x_n, y_n) pseudo-deterministically in time $2^{n+O(\epsilon n)}$. By the definition of rKt, this implies that

$$rKt(x_n, y_n) \leq n + O(\epsilon n).$$

We also have $rKt(x_n, y_n) \geq 2n - O(\epsilon n)$ by Equation (15). Therefore, we obtain $2n - O(\epsilon n) \leq n + O(\epsilon n)$, which is a contradiction for a sufficiently small constant $\epsilon > 0$. □

We observe that MrKtP is in the exponential-time variant of PP.

Proposition 5.7. MrKtP $\in$ PEXP.

Proof. Since PP is closed under truth-table reductions [26], it suffices to show that MrKtP is reducible to PP via an exponential-time truth-table reduction. Let $(x, 1^s)$ be an input such that $x \in \{0,1\}^n$ and $s \in \mathbb{N}$. For each string $d \in \{0,1\}^*$ of length at most s, we ask the PP oracle whether $U(d, r)$ outputs x in time t for at least a 2/3-fraction of $r \in \{0,1\}^t$, where $t := 2^{s-|d|}$. We accept the input $(x, 1^s)$ if and only if a positive answer is returned from the oracle. □

We expect that Theorem 5.3 can be extended to a lower bound against PP-type algorithms. However, the original motivation of Oliveira [62] is to study a gap version of MrKtP, for which the upper bound of pr-BPE can be proved. Specifically, let GapMrKtP denote the promise problem $(\Pi_{\mathrm{Yes}}, \Pi_{\mathrm{No}})$ such that $\Pi_{\mathrm{Yes}} = \{(x, s) : rKt(x) \leq s\}$ and $\Pi_{\mathrm{No}} = \{(x, s) : rKt(x) > s + c \log |x|\}$, where c is a sufficiently large constant. One can observe GapMrKtP $\in$ pr-BPE [62]. Theorem 5.3 can be extended to this promise problem. Here, abusing a notation, for a class C of languages, we say that a promise problem $\Pi \in C$ if there exists a language $L \in C$ such that $\Pi_{\mathrm{Yes}} \subseteq L \subseteq \{0,1\}^* \setminus \Pi_{\mathrm{No}}$ for $(\Pi_{\mathrm{Yes}}, \Pi_{\mathrm{No}}) = \Pi$. The lower bound of Theorem 5.8 does work against a bounded probabilistic algorithm that satisfies the promise of BPP-type algorithms over all the inputs. In contrast, the upper bound pr-BPE = pr-BPTIME($2^{O(n)}$) holds only for an algorithm that may not satisfy the promise of BPP-type algorithms for some inputs. This leaves a qualitative gap between the lower bound and the upper bound on GapMrKtP; closing this gap is an interesting open question. We note, however, that there is no gap under the plausible assumption that E $\not\subseteq$ i.o.SIZE($2^{\epsilon n}$) for some constant $\epsilon > 0$, in which case pr-BPP = P = BPP [46].

Proof of Theorem 5.8. The proof is essentially the same with Theorem 5.3. We only explain how to modify the proof of Theorem 5.3. Toward a contradiction, we assume the existence of a randomized algorithm M that computes some language L infinitely often, where L is a language consistent with GapMrKtP. For all large $n \in \mathbb{N}$ and every $x \in \{0,1\}^n$ and $s \in [n]$, we assume that the binary encoding of (x, s) is exactly equal to $m(n)$, where $m(n) = \Theta(n)$ is some function. This ensures that for infinitely many $n \in \mathbb{N}$, the algorithm M computes L correctly on input (x, s) for every $x \in \{0,1\}^n$ and every $s \in [n]$. First, we explain how to extend the lower bound of Theorem 5.3 to the infinitely-often version. Let n be the length of $x_n \in \{0,1\}^n$ and $y_n \in \{0,1\}^n$ constructed in the proof of Theorem 5.3. By inspection, in the proof of Theorem 5.3, we use the algorithm M on inputs of length at most n', where $n' = O(n)$. Any input x of length less than n' can be padded to an input of length n' by mapping x to $x' := x1^{n'-|x|}$. This ensures that the length of any input to M that we use in the proof is exactly equal to n'. Moreover, the correctness of the proof remains unchanged because $rKt(x) = rKt(x') \pm O(\log n)$. Second, we explain how to deal with the promise problem. In the proof of Theorem 5.3, we exhaustively search the lexicographically first string $y_n \in \{0,1\}^n$ such that $rKt(y_n) \geq n$. Instead, we search the lexicographically first string $y_n \in \{0,1\}^n$ such that $(y_n, n - 2c \cdot \log n) \notin L$. Since $L \subseteq \{0,1\}^* \setminus \Pi_{\mathrm{No}}$, this ensures that

$rKt(y_n) \geq n - c \log n$. Similarly, we search the lexicographically first string $x_n \in \{0,1\}^n$ such that $((x_n, y_n), 2n - O(\epsilon n)) \notin L$. This ensures that $rKt(x_n, y_n) \geq 2n - O(\epsilon n)$. Since the language L can be decided by the randomized algorithm M, we can compute the pair (x_n, y_n) pseudo-deterministically in time $2^{n+O(\epsilon n)}$. This is a contradiction. □

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